ZUDIO!

Building Zudio with Claude

A native Mac generative music app built in one week using AI-driven vibe coding

Zack Urlocker - Build To Scale - 2026

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THE INSPIRATION

A dream project - finally possible

Wanted to build a generative music app like Eno's Reflection or Jarre's EON - but lacked the coding skills. Goal: a controllable system, not a black box.

Kraftwerk | Neu! | Brian Eno | Jean-Michel Jarre

WHY MOTORIK?

Formulaic by design - perfect for rule-based generation. Apache beat, hypnotic 4/4 pulse. One chord. Locked rhythm. Infinite.

APPROACH

Research first, code second

- **Days 1-2 - Research**
  Analyzed classic Motorik tracks. Derived rules for bass, drums, leads, pads.
- **Pivot - ChatGPT to Claude**
  Claude Code took over - like going from an intern to a ten-year veteran.
- **Day 3 - Working prototype**
  MIDI tracks on screen, music playing.

> *Please carefully review the development plan and prototype docs to determine what improvements could be made to increase the likelihood of success in the programming phase.*

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Motorik Style — Research & Implementation

Context

Motorik is Zudio's original generative music style. It is derived from the krautrock and motorik groove tradition — specifically the hypnotic 4/4 pulse pioneered by Neu!, Harmonia, Kraftwerk, and their contemporaries. This document covers both the research behind the style and the complete implemented generator specification.

For a higher-level description of how Motorik fits within the app's overall architecture, see `docs/architecture.md`. The companion implementation reference for cross-style rules (tonal governance, generation pipeline, performance engineering) is `docs/implementation.md`.

Part 1: Genre Overview

Motorik (German: "motor-like") is the defining rhythmic identity of the krautrock movement. The term describes the hypnotic, mechanically precise 4/4 pulse pioneered by Klaus Dinger of Neu! — sometimes called the "Apache beat" after its influence on hip-hop. The key characteristics: pulse continuity above all else, sparse harmonic movement, modal tonality, and melodic material that is motif-first and repetition-heavy. The groove never stops; everything else orbits around it.

Research methods

Research inputs used:

- Literature and context sources on Motorik and related artists

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Bass rules

One rule selected per song. All patterns anchor beat 1 as primary attack, matching the kick drum.

Bass variation for simple rules (BAS-001/002/004): in B sections or after bar 48, substitutes a slightly more complex variant to maintain musical interest without abandoning the rule's identity:

- BAS-001 variation: quarter-note walk root→third→fifth→root (mode-correct)
- BAS-002 variation: root–root–third–root quarter pulse (mode-correct third on beat 3)
- BAS-004 variation: root (long)→third→fifth arc (mode-correct passing third)

Status log emits `BASS-EVOL` when variation begins; `BASS-DEVOL` when it reverts. Variation runs through qualifying section(s) only — alternates every other A section after bar 48.

Bass rule catalog:

- BAS-001 Root Anchor (10%): root beat 1 (long sustain), chord tone beat 3. Simplest Motorik bass.
- BAS-002 Motorik Drive (10%): four staccato quarter notes, velocity accented beats 1+3.
- BAS-003 Crawling Walk (7%): 2-bar root/fifth/approach-note pattern. Slow stepwise motion.
- BAS-004 Neu! Hallogallo lock (10%): root beat 1 (long), fifth beat 3, kick-locked.
- BAS-005 McCartney Drive (12%): 8th-note groove; bar 1 descends root→m3→5→m3, bar 2 breathes with root sustain + approach pickup. ~1/3 chance of all-drive 4-bar block.
- BAS-006 LA Woman Sustain (6%): root holds most of bar, chromatic neighbor shimmer at end.
- BAS-007 Hook Ascent (11%): high-register melodic riff; bar 1 hammers major third in 8th notes then descends, ba falls to root with minor 6th color. Inspired by Peter Hook / Joy Division.
- BAS-008 Moroder Pulse (9%): staccato 8ths root-root-fifth-fifth-b7-b7-root-root. Mechanical, relentless. Inspired by Giorgio Moroder "I Feel Love".
- BAS-009 Vitamin Hook (7%): bar 1 climbs root→fifth→octave with passing tone, bar 2 descends and breathes.

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Generation Pipeline

10-step pipeline

```
1.  Musical Frame     — key, mode, tempo, mood, total bars, loop lengths
2.  Song Structure    — sections (intro/A/B/bridge/outro), chord plan
3.  Tonal Map         — per-bar chord windows with note pools for each track role
4.  Per-track loops   — each generator produces its event sequence
5.  Loop tiling       — loops tiled to full song length (Ambient); or full-song
                        generation (Kosmic/Motorik)
6.  Post-processing   — dynamic arc, density gating, harmonic filtering
7.  Step annotations  — bar markers for status log playback feed
8.  Generation log    — rule IDs written to status box and MIDI log file
9.  SongState         — all of the above packaged into an immutable struct
10. Playback load     — SongState handed to PlaybackEngine; pre-computes note-off map
```

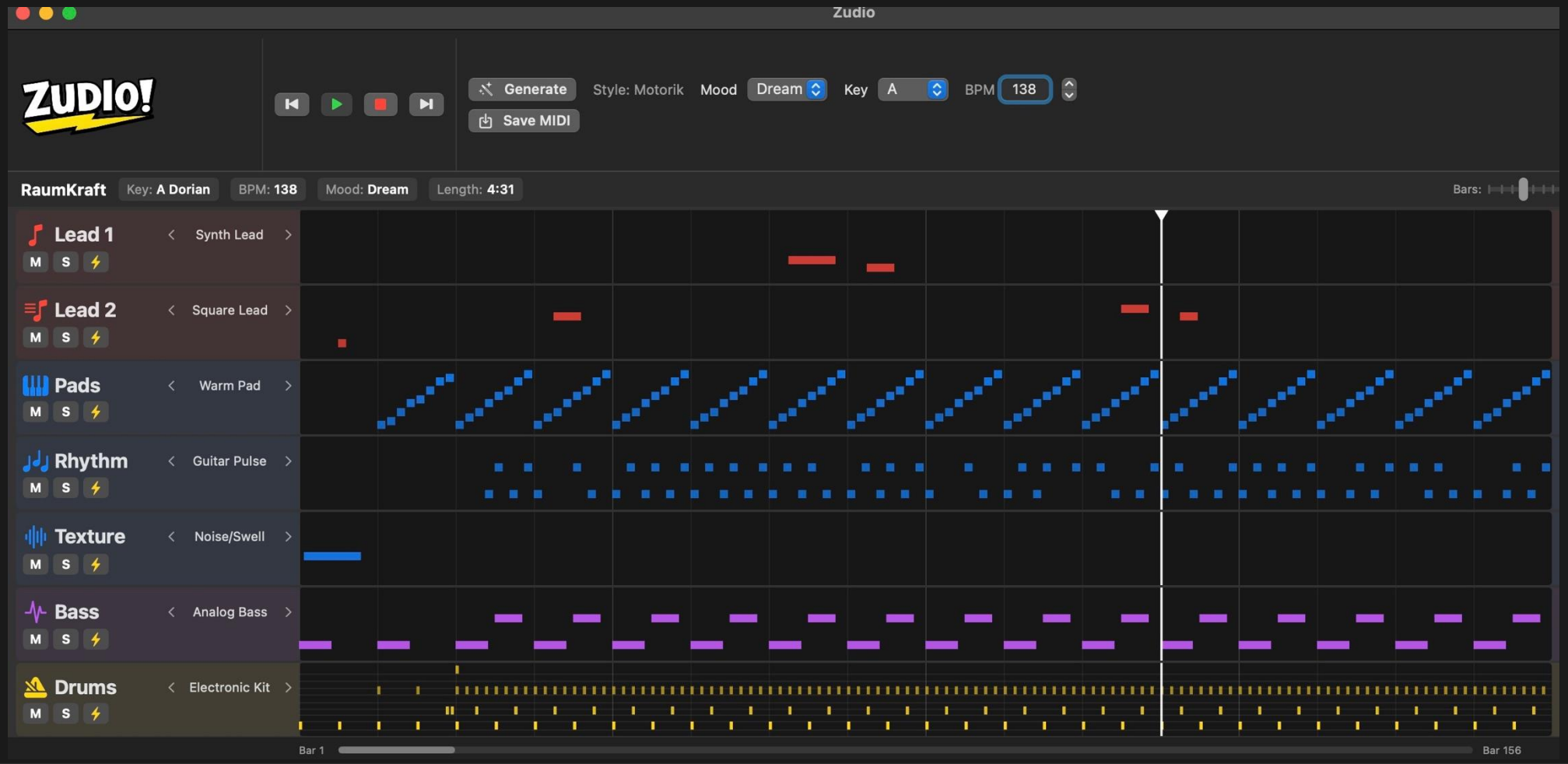
Total bars is derived from a triangular distribution: min 210s (3:30), peak 285s (4:45), max 390s (6:30). Rounded to nearest multiple of 4 bars: `totalBars = round((seconds × tempo) / 60 / 4) × 4`. All tracks must generate exactly `totalBars` bars of MIDI events.

Seeded RNG (SplitMix64)

Same seed + same controls = same song. Per-track seed: `splitmix64(globalSeed XOR (trackIndex × 0x9e3779b97f4a7c15))`. Per-track regenerate generates a new entropy value for that track only.

The global seed and any per-track override seeds are written to the `.txt` log file saved alongside each MIDI export. **Load**

Piano roll, MIDI tracks, generation log



KEY INSIGHT

AI cannot hear - you can

Claude found a hardcoded bass riff that clashed in certain keys. It wrote consistent code for one mode and never checked the others.

I caught it because something bothered me. The AI had no way to know it should be bothered.

> *Nobody checked! The code assumed the song would always be in a minor mode.*

WHAT AI DOES WELL

Explains theory - finds bugs when directed - generates good code

WHAT AI CANNOT DO YET

Notice something sounds wrong - exercise taste

WHAT YOU BRING

Taste - curiosity - judgment

MUSICAL STYLES

Three genres, one engine

MOTORIK

Kraftwerk - Neu! - Hallogallo

Apache beat, locked bass, hypnotic pulse. 3 days to build.

1 chord | 4/4 pulse

KOSMIC

Tangerine Dream - Jean-Michel Jarre

Evolving synth textures, distinct chord structures. About 2 days.

Arpeggios | Pads

AMBIENT

Brian Eno - ECM Jazz

Variable-length tape loops on MIDI. Shakers, bongos, sparse feel. A few hours.

AMB-DRUM-004 Claude hand percussion - shaker-as-pulse with sparse conga syncopation. Brazilian/Afro-Cuban feel, ECM-style ambient jazz.

QUALITY LOOP

Using AI to test AI

Fed generated songs back to Claude for analysis. It found more errors than I did - tonal clashes, instruments stepping over each other, notes not adjusted to scale.

Generate song → Claude analyzes → Fix rules

↻

> *The b7 is +10 in Dorian but +11 in Phrygian - so +10 blindly applied is wrong when the mode is not minor.*

I had no idea if this was true - but it made sense, and it worked.

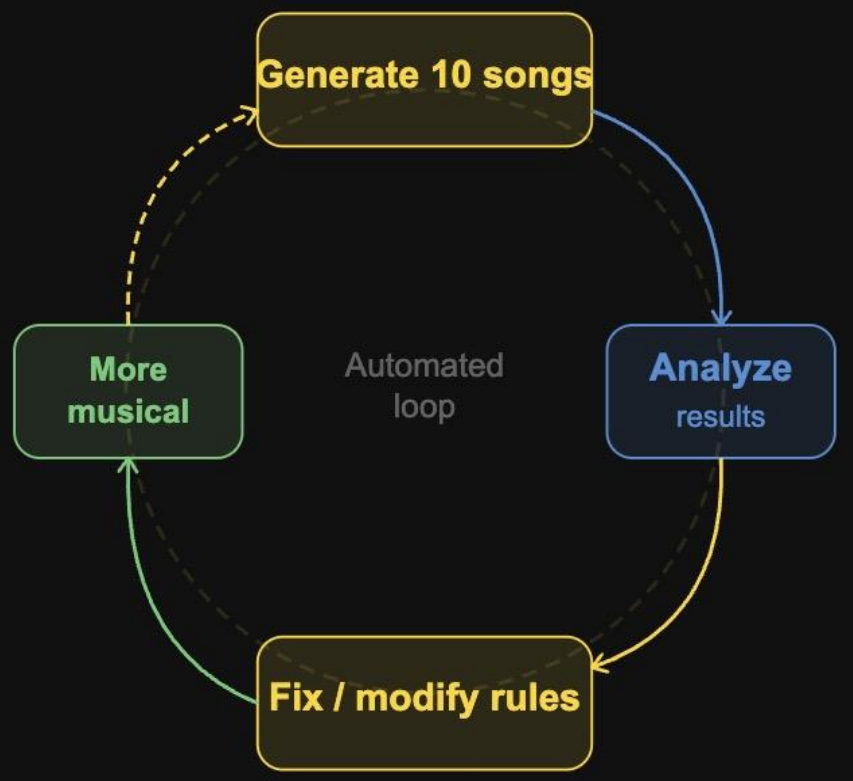
NEW STYLE

Chill - automated quality loop

Chill-out jazz influenced by Moby, St Germain, Electric Buddha Band.

This time: Claude ran the full QA test loop itself - generate, analyze, refine.

Faster convergence | Higher quality | Fewer tonal clashes



RESULTS

One week of vibe coding

15-30x
productivity gain vs. human effort

1 wk
finished app vs. 6-9 months

Generates listenable music across three styles.

Tracks export to MIDI for refinement in Logic Pro.

Songs regenerated track-by-track.

github.com/ZUrlocker1/Zudio

Swift / SwiftUI | CoreMIDI | Claude Code | Open source

TAKEAWAYS

What vibe coding taught me

SPECS ARE EVERYTHING

AI amplifies your prep work - good or bad. Plan first.

AI AS PARTNER

The breakthrough comes when you work with AI as a collaborator, not a tool.

JUDGMENT WINS

What cannot be automated is knowing when something is genuinely good.

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